

IKEY, Russian Federation

WMO: 305070

Lat: 54.183N

Lon: 100.083E

Elev: 1729

StdP: 13.80

Time Zone: 8.00 (E08)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.1	-30.0	-39.3	0.6	-36.0	-32.9	0.9	-29.8	21.7	7.4	12.2	4.3	0.4	320	0.636

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	80.7	64.2	77.5	63.1	74.5	62.0	67.2	76.5	65.2	74.5	63.4	72.1	6.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.7	94.0	72.9	61.5	87.0	70.5	59.7	81.7	68.2	32.8	76.7	31.2	74.6	29.7	72.3	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.1	12.2	9.7	-41.8	86.7	5.2	3.8	-45.6	89.4	-48.6	91.6	-51.5	93.8	-55.3	96.6		
(5)			-42.2	69.8	4.9	2.1	-45.8	71.3	-48.7	72.5	-51.4	73.7	-55.0	75.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	30.1	-5.1	0.6	15.6	33.0	47.2	56.5	61.9	58.2	46.3	32.9	13.3	-0.7	(6)
(7)		DBStd	25.23	12.26	11.45	10.60	8.10	7.67	6.52	5.30	4.94	6.44	7.76	12.01	12.46	(7)
(8)		HDD50	8190	1708	1384	1067	512	147	18	2	3	143	532	1102	1572	(8)
(9)		HDD65	12766	2173	1804	1532	961	552	262	121	214	561	997	1552	2037	(9)
(10)		CDD50	933	0	0	0	0	60	212	371	258	32	0	0	0	(10)
(11)		CDD65	36	0	0	0	0	0	7	25	4	0	0	0	0	(11)
(12)		CDH74	737	0	0	0	0	75	218	357	81	6	0	0	0	(12)
(13)		CDH80	122	0	0	0	0	13	41	59	9	0	0	0	0	(13)
(14)	Wind	WSAvg	3.0	1.9	2.1	3.2	4.7	4.4	3.8	3.1	2.8	3.0	2.7	2.5	1.7	(14)
(15)	Precipitation	PrecAvg	17.0	0.5	0.4	0.4	0.7	1.4	2.6	3.3	3.3	2.0	0.9	0.7	0.6	(15)
(16)		PrecMax	25.4	0.9	0.7	0.9	1.9	2.7	6.8	6.4	6.3	4.3	2.4	1.6	1.1	(16)
(17)		PrecMin	10.7	0.1	0.2	0.1	0.2	0.7	1.1	1.0	1.1	0.7	0.2	0.4	0.2	(17)
(18)		PrecStd	3.3	0.2	0.2	0.2	0.4	0.5	1.3	1.2	1.1	0.9	0.5	0.4	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.9	36.7	49.8	63.4	81.6	84.6	85.4	80.1	74.2	63.4	45.3	34.3	(19)
(20)			MCWB	26.8	31.2	40.3	47.2	58.6	63.9	68.4	65.8	58.9	50.0	38.4	29.1	(20)
(21)		2%	DB	21.4	28.1	42.6	57.3	75.3	79.9	81.5	75.6	68.3	55.0	36.9	26.2	(21)
(22)			MCWB	18.9	23.9	34.7	44.2	55.8	61.8	66.7	63.2	57.5	45.9	31.9	23.4	(22)
(23)		5%	DB	16.3	23.0	37.9	52.3	70.4	76.5	78.0	73.2	63.3	49.7	32.3	21.1	(23)
(24)			MCWB	14.7	19.6	31.5	41.8	52.8	61.7	65.2	62.2	54.5	42.6	28.8	18.8	(24)
(25)		10%	DB	11.9	18.4	33.8	47.1	65.4	71.9	75.1	70.3	58.7	45.2	28.9	17.4	(25)
(26)			MCWB	10.9	15.9	28.9	38.4	50.8	59.0	64.4	61.2	51.8	39.6	26.3	15.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.9	31.7	41.0	49.0	60.2	67.0	71.2	68.3	61.5	51.6	38.1	29.0	(27)
(28)			MCDWB	29.7	37.3	49.0	60.3	78.3	77.3	82.0	76.5	70.1	61.1	45.0	33.5	(28)
(29)		2%	WB	19.0	24.0	35.2	45.2	56.8	64.5	68.4	65.3	57.8	46.6	32.3	23.4	(29)
(30)			MCDWB	20.9	27.9	42.1	56.7	72.9	77.0	78.1	73.1	66.6	54.0	36.5	26.1	(30)
(31)		5%	WB	14.6	19.9	31.8	42.4	54.3	62.3	66.6	63.2	55.3	43.2	28.8	19.4	(31)
(32)			MCDWB	16.0	23.0	37.4	51.8	67.8	73.9	76.1	71.0	61.6	48.6	31.7	21.3	(32)
(33)		10%	WB	10.8	16.0	28.8	38.8	51.5	60.0	64.8	61.5	52.6	40.0	26.4	15.9	(33)
(34)			MCDWB	11.8	18.3	33.8	46.5	63.4	70.0	73.6	68.8	58.4	44.7	28.9	17.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	20.3	26.3	26.5	20.1	24.9	22.1	20.4	19.5	20.6	18.8	18.8	19.1	(35)
(36)			MCDBR	22.3	30.0	31.2	30.0	37.5	32.7	27.5	27.3	30.7	28.5	20.2	23.0	(36)
(37)			MCWBR	20.2	25.9	24.1	19.2	21.3	17.2	15.2	16.1	19.3	19.8	15.9	20.0	(37)
(38)		5% WB	MCDWB	21.7	29.3	29.9	28.3	33.9	27.2	24.5	24.5	27.8	26.8	18.9	21.9	(38)
(39)			MCWBR	19.8	25.5	23.1	18.7	20.2	15.4	14.7	15.1	18.4	19.2	15.2	19.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.271	0.304	0.343	0.381	0.421	0.417	0.417	0.390	0.345	0.325	0.300	0.274	(40)	
(41)		taud	2.087	2.064	2.084	2.204	2.138	2.258	2.291	2.345	2.423	2.354	2.181	2.047	(41)	
(42)		Ebn at Noon	209	242	259	267	261	264	262	262	261	238	196	176	(42)	
(43)		Edh at Noon	22	31	39	40	45	41	38	34	28	24	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	327	650	1135	1498	1660	1834	1662	1418	1010	600	319	223	(44)	
(45)		RadStd	21	38	57	92	144	108	145	100	92	46	30	20	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page